

Lipschitz Equivalent Separable Banach Spaces Need Not Be Linearly Isomorphic

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Abstract

There are separable real Banach spaces that are globally bi-Lipschitz equivalent but not linearly isomorphic. This gives a negative answer to the separable Banach-space Lipschitz-isomorphism problem.

1 Introduction

A Banach-space norm defines both a linear topology and a metric. A bounded linear isomorphism preserves the metric up to multiplicative constants, but a map with that metric property need not preserve addition or scalar multiplication. The Lipschitz-isomorphism problem asks whether a *bijection* with two-sided Lipschitz bounds nevertheless forces two Banach spaces to be linearly isomorphic. We consider the separable real case and obtain a negative answer.

Throughout, ℓ_2 denotes the real separable Hilbert space of square-summable scalar sequences. For a Banach space H , set

$$c_0(H) = \{(v_i)_{i \geq 1} : v_i \in H, \|v_i\| \rightarrow 0\}, \quad \|(v_i)\| = \sup_i \|v_i\|.$$

Ordinary c_0 means $c_0(\mathbb{R})$. A linear embedding means a bounded linear map that is bounded below; its range is consequently closed.

Theorem 1.1. *There are separable real Banach spaces X, Y and a bijection $\Psi : X \rightarrow Y$ such that, for every $s, t \in X$,*

$$\frac{4}{21} \|s - t\|_X \leq \|\Psi(s) - \Psi(t)\|_Y \leq \frac{76}{25} \|s - t\|_X. \quad (1.1)$$

The space X contains a linearly isometric copy of $c_0(\ell_2)$, while Y contains no closed subspace linearly isomorphic to $c_0(\ell_2)$. In particular, X and Y are not linearly isomorphic.

The obstruction in this theorem is deliberately block-valued: each coordinate of $c_0(\ell_2)$ is an entire Hilbert space. It does not say that Y contains no ordinary c_0 . The Hilbert block in each coordinate is essential to the obstruction proved below.

Historical setting

Ribe proved that uniformly homeomorphic normed spaces have the same finite-dimensional linear structure up to a uniform distortion bound [14, Theorem 1]. That result shows how much linear information can survive even a weaker form of metric equivalence. At the same time, Aharoni and Lindenstrauss constructed Lipschitz equivalent, nonisomorphic

nonseparable Banach spaces and explicitly asked for separable examples [1, pp. 281–282]. Ribe later constructed separable uniformly homeomorphic, nonisomorphic spaces [15]. Aharoni and Lindenstrauss then gave uniform pairs that are not Lipschitz equivalent [2], while Johnson, Lindenstrauss and Schechtman developed further separable uniform examples and left the separable Lipschitz question open [9, Theorem 5.8 and Section 8(3)]. These developments make the global two-sided estimate in (1.1) a substantive requirement, rather than a reformulation of a uniform counterexample.

Several positive results explain why the separable question resists a direct transfer of those examples. Godefroy, Kalton and Lancien proved that a Banach space Lipschitz isomorphic to c_0 is linearly isomorphic to c_0 , and that being linearly isomorphic to a subspace of c_0 is invariant under Lipschitz isomorphism [8, Theorems 2.1–2.2]. Godefroy and Kalton’s lifting theorem places every separable Banach space linearly and isometrically inside its Lipschitz-free space; their quotient corollary converts a Lipschitz right inverse of a linear quotient onto a separable space into a bounded linear right inverse [7, Theorem 3.1 and Corollary 3.2]. The construction below uses free spaces; Remark 3.2 explains the distinction between the lift used here and a right inverse of the linearized correction.

In the quasi-Banach setting, Albiac and Kalton obtained separable counterexamples for every $0 < p < 1$ [4, Theorem 5.4]. The strict inequality $p < 1$ is part of that theorem, so it does not settle the Banach-norm case. Kalton recorded the separable Lipschitz-isomorphism question as Problem 3 in his 2008 survey [11, Section 2.2]; Albiac, Ansorena, Bóna and Cúth still described the Banach case as open in the first paragraph of their paper published online in December 2025 [3]. Theorem 1.1 resolves that question negatively for separable real Banach spaces. It asserts neither reflexivity nor a complex-scalar analogue.

For comparison, every surjective isometry between the unit spheres of nonzero real Banach spaces extends uniquely to a surjective real-linear isometry [13, Theorem 1.1]. That result concerns exact distances on unit spheres; here we allow fixed multiplicative distortion but require a bijection of the whole spaces.

Proof overview

The proof separates the metric construction from the linear obstruction. Its intermediate map acts between Hilbert spaces:

$$M = U \oplus_2 V, \quad V = \ell_2, \quad h : M \longrightarrow U.$$

Here $U = (\bigoplus_{n \geq 1} U_n)_2$ is a Hilbert sum of finite-dimensional coordinate blocks. The map h is an onto bi-Lipschitz change of variables fixing zero, close to an isometry. Its relevant nonlinear part is

$$D(u, v) = h(u, v) - u.$$

The construction arranges that, near any finite family of distinct points of M , the sufficiently high output coordinates of D lie in mutually orthogonal coordinate subspaces. The cutoff index and the neighborhoods may depend on that finite family. We call this property *local orthogonality*.

To turn this geometry into a linear obstruction, use the Lipschitz-free space $E = \mathcal{F}(M)$, the closed linear span of evaluation functionals on real zero-preserving Lipschitz functions. The evaluation map $\delta : M \rightarrow E$ preserves distances. Every zero-preserving Lipschitz map has a bounded linearization, so D determines $q : E \rightarrow U$ by

$$q\delta(u, v) = h(u, v) - u.$$

Section 4 proves that local orthogonality makes q *completely continuous*: a weakly null sequence in E has norm-null image. This is the one place where the local coordinate geometry enters the Banach-space obstruction.

The linearization is also what makes the metric change of variables possible. Give $E \oplus V$ its sum norm and put

$$Q_1(e, v) = (qe, v), \quad B(e, v) = e + \delta(qe, v).$$

The identity $qB = hQ_1$ is triangular: applying q to the new variable records h of the old output. Given $z \in E$, one first recovers $x = h^{-1}(qz)$ and then subtracts $\delta(x)$; this yields the explicit inverse of B . Section 3 proves both inverse identities and keeps track of the sum norm.

For a bounded linear $Q : A \rightarrow H$, the graph space used in the final step consists of sequences $s = (s_i)$ for which $\sum_i 2^{-i} \|s_i\| < \infty$ and $(Qs_i) \in c_0(H)$, with norm

$$\|s\|_{Z_Q} = \max \left\{ \sum_i 2^{-i} \|s_i\|, \sup_i \|Qs_i\| \right\}.$$

Set $X = Z_{Q_1}$ and $Y = Z_q$. Applying B in each coordinate gives the desired bijection: $qB = hQ_1$ controls the second part of the graph norm, and the Lipschitz bounds for B control the weighted sum. The V coordinates embed $c_0(V)$ isometrically in X . By contrast, complete continuity of q makes each fixed output coordinate tend to zero along an orthonormal sequence in any Hilbert block of a proposed $c_0(\ell_2)$ copy in Y . A disjoint-support argument then finds a unit vector in each block with arbitrarily small full $c_0(U)$ output. A surviving copy would embed ordinary c_0 into $\ell_1(E)$. Section 2 excludes this because $E = \mathcal{F}(M)$ is weakly sequentially complete.

Two analytic steps supply the inputs just described. For localized linearization, a hypothetical non-null image of a weakly null sequence produces uniformly Lipschitz scalar tests d_i . Orthogonality allows only finitely many points where their limiting local Lipschitz constants remain large. Cutoffs remove neighborhoods of those points; a shrinking-ball lemma shows that the removed tests pair negligibly with the weakly null sequence. The compact reduction theorem of Aliaga, Noûs, Petitjean and Procházka [5, Theorem 2.3] then reduces the remainder to a compact set, where a finite cover controls its Lipschitz constant.

For the Hilbert map, a unit vector in each block is designated as a slot. With the slots frozen, perpendicular components stay in their blocks, old slot coordinates move to even blocks, and the additional ℓ_2 coordinates fill odd blocks. This is an onto linear isometry. The slots are then allowed to depend on the input so that near different points they eventually use disjoint coordinates. Finite approximants freeze all but finitely many slots and have uniform global inverse bounds. Section 5 constructs the slots with a common radial derivative budget, proves global invertibility of the finite approximants, and passes to an onto limit. Its decisive estimate bounds every tail of the inverse images by the corresponding tail of a fixed output, with an exact cancellation of slot terms. Section 6 combines the results and computes the constants in (1.1).

2 Free spaces and the graph-space obstruction

The metric construction will provide a bounded linear operator $q : E \rightarrow U$ that is completely continuous, with E weakly sequentially complete. This section explains why those two properties exclude $c_0(\ell_2)$ from the associated graph space. We first record the free-space facts that produce E , then prove the obstruction without assuming compactness of any operator into $c_0(U)$.

2.1 Free-space facts and the two external inputs

For a pointed metric space $(M, d, 0)$, let $\text{Lip}_0(M)$ be the Banach space of real Lipschitz functions $f : M \rightarrow \mathbb{R}$ with $f(0) = 0$, normed by $\text{Lip}(f)$. Let $\delta(x) \in \text{Lip}_0(M)^*$ be evaluation

at x , and define

$$\mathcal{F}(M) = \overline{\text{span}}\{\delta(x) : x \in M\} \subset \text{Lip}_0(M)^*.$$

The evaluation and molecule construction goes back to Arens and Eells [6, Sections 2–3]; the notation here is the later Lipschitz-free formulation. In particular, $\delta(0) = 0$. When a Lipschitz function f need not vanish at 0, the pairing $\langle f, \mu \rangle$ means $\langle f - f(0), \mu \rangle$.

The evaluation map preserves distances:

$$\|\delta(x) - \delta(y)\| = d(x, y). \quad (2.1)$$

The Lipschitz inequality gives the upper bound. For the reverse bound, test against $t \mapsto d(t, y) - d(0, y)$, which has Lipschitz constant at most one. If M is separable, choose a countable dense set $D_0 \subset M$. Equation (2.1) shows that finite rational linear combinations of $\delta(D_0)$ are dense in $\mathcal{F}(M)$; hence $\mathcal{F}(M)$ is separable.

If $K \subset M$ contains 0, the closed span of $\delta(K)$ in $\mathcal{F}(M)$ is isometric to $\mathcal{F}(K)$. To see this, a real L -Lipschitz function f on K extends with the same Lipschitz constant by the McShane formula [10, Lemma 3.1 and the following paragraph]

$$\tilde{f}(x) = \inf_{z \in K} (f(z) + L d(x, z)).$$

For $L = 0$ use the constant extension. For $L > 0$, the Lipschitz inequality on K makes the displayed infimum equal to f there, and the triangle inequality gives the L -Lipschitz bound on M . The infimum is finite because $0 \in K$ supplies an upper bound and the triangle inequality supplies a lower bound. Thus the suprema defining the two free-space norms agree on finite evaluation sums supported in K , and then on their closures. In particular,

$$|\langle f, \nu \rangle| \leq \text{Lip}(f|_K) \|\nu\| \quad (\nu \in \mathcal{F}(K)). \quad (2.2)$$

For any Banach space H and any Lipschitz map $D : M \rightarrow H$ with $D(0) = 0$, there is a unique bounded linear map

$$\hat{D} : \mathcal{F}(M) \rightarrow H, \quad \hat{D}\delta(x) = D(x), \quad \|\hat{D}\| = \text{Lip}(D). \quad (2.3)$$

Indeed, for a finite evaluation sum, define $\hat{D}(\sum_j a_j \delta(x_j)) = \sum_j a_j D(x_j)$. Testing the latter vector against a norm-one functional in H^* and using the free-space norm gives

$$\left\| \sum_j a_j D(x_j) \right\| \leq \text{Lip}(D) \left\| \sum_j a_j \delta(x_j) \right\|.$$

This inequality makes the definition independent of the chosen representation and extends it continuously to $\mathcal{F}(M)$. Testing the extension on $\delta(x) - \delta(y)$ proves the reverse norm bound. This metric-domain linearization is recorded in [10, Lemma 3.2]; for Banach-space domains see also Godefroy and Kalton [7, Lemma 2.5].

A bounded linear operator is *completely continuous* if it sends each weakly null sequence to a norm-null sequence. A Banach space is *weakly sequentially complete* if each weakly Cauchy sequence has a weak limit in the space. We use the following two external results with exactly the displayed hypotheses.

Theorem 2.1 (Compact reduction for a weakly null sequence). *Let $(M, d, 0)$ be a complete pointed metric space, and let $\mu_i \rightarrow 0$ weakly in $\mathcal{F}(M)$. For every $\tau > 0$, there is a compact $K \subset M$, with $0 \in K$, such that*

$$\text{dist}(\mu_i, \mathcal{F}(K)) < \tau \quad \text{for every } i.$$

This is the sequential consequence of the tightness theorem of Aliaga, Noûs, Petitjean and Procházka [5, Theorem 2.3, arXiv version cited in the bibliography]. Their theorem states that every weakly precompact subset W of $\mathcal{F}(M)$, for complete M , is tight: for each $\epsilon > 0$ there is compact $K_0 \subset M$ with $W \subset \mathcal{F}(K_0) + \epsilon B_{\mathcal{F}(M)}$, where for an arbitrary subset K_0 this notation means its closed evaluation span inside $\mathcal{F}(M)$. Here weak precompactness means that every sequence in W has a weakly Cauchy subsequence. The set $\{0, \mu_1, \mu_2, \dots\}$ has that property: any sequence in it has either a constant subsequence or a subsequence whose indices tend to infinity, which converges weakly to zero. Apply the theorem with $\epsilon = \tau/2$ and enlarge K_0 by 0. This gives the strict inequality above. Kalton's finite-radius localization lemma [10, Lemma 4.5] is a precursor to this complete-space compact reduction.

Theorem 2.2 (Weak sequential completeness of a free space). *If M is a superreflexive real Banach space, then $\mathcal{F}(M)$ is weakly sequentially complete. In particular, this applies to every real Hilbert space M .*

This is [5, Corollary 2.5, arXiv version cited in the bibliography]. Its compact-set input is the Main Theorem of Kochanek and Pernecká [12], which concerns free spaces over compact subsets of superreflexive Banach spaces, not directly the whole Hilbert space.

2.2 Weighted graph spaces

Let A, H be real Banach spaces and $Q : A \rightarrow H$ bounded and linear. Define

$$Z_Q = \{s = (s_i)_{i \geq 1} : s_i \in A, \sum_{i \geq 1} 2^{-i} \|s_i\| < \infty, (Qs_i)_{i \geq 1} \in c_0(H)\}, \quad (2.4)$$

$$\|s\|_{Z_Q} = \max \left\{ \sum_{i \geq 1} 2^{-i} \|s_i\|, \sup_{i \geq 1} \|Qs_i\| \right\}. \quad (2.5)$$

The membership condition is part of the definition; finiteness of the displayed maximum alone would not require the Q -images to tend to zero.

Lemma 2.3. *Z_Q is a Banach space, and it is separable if A, H are separable. The coordinate map $p_i : Z_Q \rightarrow A$, $p_i(s) = s_i$, has norm at most 2^i , and*

$$J_Q : Z_Q \rightarrow c_0(H), \quad J_Q(s) = (Qs_i)_i,$$

has norm at most one.

Proof. Let $\ell_1(2^{-i}; A)$ denote the weighted sum space with norm $\sum_i 2^{-i} \|s_i\|$. The map $s \mapsto (2^{-i} s_i)_i$ is an isometry from this space onto $\ell_1(A)$, so it is complete. The map

$$s \mapsto (s, (Qs_i)_i)$$

identifies Z_Q isometrically with a subspace of $\ell_1(2^{-i}; A) \oplus_\infty c_0(H)$. This subspace is closed. In fact, suppose $(s^{(k)}, (Qs_i^{(k)})_i)$ converges there to (s, y) . For each fixed i ,

$$\|s_i^{(k)} - s_i\| \leq 2^i \|s^{(k)} - s\|_{\ell_1(2^{-j}; A)} \longrightarrow 0.$$

Continuity of Q gives $Qs_i^{(k)} \rightarrow Qs_i$, whereas convergence in $c_0(H)$ gives $Qs_i^{(k)} \rightarrow y_i$. Thus $y_i = Qs_i$ for all i , as required. Completeness follows.

If A, H are separable, truncating a sequence in Z_Q gives convergence in both parts of (2.5): the weighted sum has a vanishing tail, and the Q -image is norm-null. Choose a fixed countable dense subset of A . The finitely supported sequences with entries in that subset form a countable dense subset of Z_Q , because each of their finitely many entries may be approximated in A and Q is continuous. Finally, the two claimed operator bounds are immediate from (2.5). $\square$

The next lemma records exactly the consequence of weak sequential completeness that the graph obstruction needs. It does not assert that every operator into a c_0 -sum is compact.

Lemma 2.4. *If E is a weakly sequentially complete real Banach space, then $\ell_1(E)$ contains no closed subspace linearly isomorphic to ordinary c_0 .*

Proof. Every finite direct sum of E is weakly sequentially complete: the coordinates of a weakly Cauchy sequence are weakly Cauchy in E , hence have weak limits, and the finite list of limits is the weak limit in the sum. The property also passes to a closed linear subspace F of a weakly sequentially complete space G . Indeed, a weakly Cauchy sequence in F is weakly Cauchy in G ; its weak limit lies in F , since a closed linear subspace is weakly closed by Hahn–Banach separation. Every functional on F extends to G by Hahn–Banach, so the same limit is weak in F .

Ordinary c_0 is not weakly sequentially complete. If (e_k) is its usual unit basis, the partial sums $e_1 + \dots + e_n$ are weakly Cauchy: for every $(a_k) \in c_0^* = \ell_1$, their pairings are the Cauchy partial sums $\sum_{k \leq n} a_k$. A weak limit would have every coordinate equal to one, which is not an element of c_0 .

Suppose now that $A : c_0 \rightarrow \ell_1(E)$ is a linear embedding with $\|Ax\| \geq b\|x\|$, where $b > 0$. Let P_N be projection onto the first N coordinates, with $P_0 = 0$. No $P_N A$ is bounded below, because otherwise its closed range would be a copy of c_0 in the weakly sequentially complete finite sum E^N .

Starting with $N_0 = 0$, choose unit vectors $x_j \in c_0$ and integers $N_j > N_{j-1}$ recursively so that

$$\|P_{N_{j-1}} Ax_j\| < b/8, \quad \|(I - P_{N_j})Ax_j\| < b/8.$$

The first inequality is possible because $P_{N_{j-1}} A$ is not bounded below; after choosing x_j , its $\ell_1(E)$ tail gives the second. Put $z_j = (P_{N_j} - P_{N_{j-1}})Ax_j$. Then

$$\|Ax_j - z_j\| < b/4, \quad \|z_j\| > 3b/4.$$

The z_j have disjoint coordinate supports. For any finite scalar family (α_j) ,

$$\begin{aligned} \left\| \sum_j \alpha_j Ax_j \right\| &\geq \left\| \sum_j \alpha_j z_j \right\| - \sum_j |\alpha_j| \|Ax_j - z_j\| \\ &= \sum_j |\alpha_j| \|z_j\| - \sum_j |\alpha_j| \|Ax_j - z_j\| \geq \frac{b}{2} \sum_j |\alpha_j|. \end{aligned}$$

Consequently,

$$\frac{b}{2\|A\|} \sum_j |\alpha_j| \leq \left\| \sum_j \alpha_j x_j \right\|_{c_0} \leq \sum_j |\alpha_j|.$$

The finite-sum map therefore extends by completeness to a bounded linear embedding of ℓ_1 into c_0 , with closed range. This is impossible. If F is any closed subspace of c_0 , the restriction map $c_0^* = \ell_1 \rightarrow F^*$ is onto by Hahn–Banach, so F^* is separable. A subspace isomorphic to ℓ_1 would instead have dual isomorphic to ℓ_∞ , which is nonseparable: the uncountable set of all 0-1 sequences is pairwise separated by distance one. This contradiction proves the lemma. $\square$

Lemma 2.5. *Let E, U be real Banach spaces, with E weakly sequentially complete, and let $q : E \rightarrow U$ be a completely continuous bounded linear operator. Then Z_q contains no closed subspace linearly isomorphic to $c_0(\ell_2)$.*

Proof. Suppose that $S : c_0(\ell_2) \rightarrow Z_q$ is a linear embedding with $\|Sx\|_{Z_q} \geq b_0\|x\|$, $b_0 > 0$. We will select a scalar copy of c_0 on which the $c_0(U)$ part of S has norm at most $b_0/2$. The weighted sum part of the graph norm must then retain the lower bound b_0 , giving a copy of c_0 in $\ell_1(E)$, contrary to Lemma 2.4.

Write $I_n : \ell_2 \rightarrow c_0(\ell_2)$ for injection into the n th coordinate and put

$$T_n = J_q S I_n : \ell_2 \rightarrow c_0(U).$$

We first prove that each T_n fails to be bounded below.

Fix n and an orthonormal sequence (e_k) in ℓ_2 . It is weakly null, since the scalar coefficients of a fixed Hilbert vector against an orthonormal sequence are square summable. For each fixed i , boundedness of $p_i S I_n$ therefore gives $p_i S I_n e_k \rightarrow 0$ weakly in E . Complete continuity of q implies

$$\|(T_n e_k)_i\| = \|q p_i S I_n e_k\| \longrightarrow 0 \quad \text{for each fixed } i.$$

Set $y_k = T_n e_k$, so $\|y_k\| \leq \|T_n\|$. Choose k_j increasing and integers $0 = m_0 < m_1 < m_2 < \dots$ as follows. Having chosen m_{j-1} , choose k_j so large that

$$\max_{i \leq m_{j-1}} \|(y_{k_j})_i\| < 2^{-j}.$$

The maximum over no coordinates is zero. Then choose $m_j > m_{j-1}$ so large that $\sup_{i > m_j} \|(y_{k_j})_i\| < 2^{-j}$, using $y_{k_j} \in c_0(U)$. Let z_j be the restriction of y_{k_j} to coordinates $m_{j-1} < i \leq m_j$. Thus

$$\|y_{k_j} - z_j\| < 2^{-j}, \quad \|z_j\| \leq \|T_n\|.$$

Because the z_j have disjoint coordinate supports, for all N

$$\left\| \sum_{j=1}^N T_n e_{k_j} \right\| \leq \left\| \sum_{j=1}^N z_j \right\| + \sum_{j=1}^N 2^{-j} \leq \|T_n\| + 1.$$

The corresponding input sum has Hilbert norm $\sqrt{N}$. No positive lower bound for T_n is compatible with these two estimates.

Choose a unit vector $t_n \in \ell_2$ such that $\|T_n t_n\| \leq b_0 2^{-n-1}$. The map

$$R : c_0 \rightarrow c_0(\ell_2), \quad R(a) = (a_n t_n)_n$$

is a linear isometry. For finitely supported a ,

$$\|J_q S R a\| \leq \sum_n |a_n| \|T_n t_n\| \leq \frac{b_0}{2} \|a\|_\infty.$$

Truncation in c_0 and continuity extend this estimate to every $a \in c_0$. On the other hand $\|S R a\|_{Z_q} \geq b_0 \|a\|_\infty$. The second part of the maximum in (2.5) is at most $(b_0/2) \|a\|_\infty$, so the first part must satisfy

$$\sum_i 2^{-i} \|(S R a)_i\| \geq b_0 \|a\|_\infty.$$

For $a = 0$ this is immediate; for $a \neq 0$ it follows because the second part is strictly below the required lower bound for the maximum. Hence

$$a \longmapsto (2^{-i} (S R a)_i)_i$$

is a bounded linear embedding of c_0 into $\ell_1(E)$. Boundedness comes from $\sum_i 2^{-i} \|(S R a)_i\| \leq \|S R a\|_{Z_q}$, and the displayed lower bound makes the range closed. This contradicts Lemma 2.4. $\square$

3 A triangular change of variables

An onto bi-Lipschitz Hilbert-space change of variables h will produce a bi-Lipschitz equivalence between two weighted graph spaces. The triangular map below has an explicit inverse obtained from h^{-1} . The identities involving h and h^{-1} control the c_0 -output, while direct Lipschitz estimates control the weighted sum. Complete continuity of the linearization of $h - \pi_U$ enters only at the final step: the preceding obstruction then shows that the two graph spaces are not linearly isomorphic.

Triangular changes of variables also appear in the quotient-section construction of Aharoni and Lindenstrauss [1, pp. 281–282] and in the free-space splitting of Godefroy and Kalton [7, Theorem 2.12]. Here the argument uses the locally verified identity $qB = hQ_1$.

Proposition 3.1 (Weighted graph criterion). *Let U be a separable real Hilbert space, $V = \ell_2$, and $M = U \oplus_2 V$, with projections π_U, π_V . Suppose $h : M \rightarrow U$ is a bijection with $h(0) = 0$ and, for all $x, x' \in M$,*

$$a\|x - x'\| \leq \|h(x) - h(x')\| \leq b\|x - x'\|, \quad 0 < a \leq b < \infty.$$

Let $E = \mathcal{F}(M)$ and let $q : E \rightarrow U$ be the linearization of $D = h - \pi_U$. Suppose q is completely continuous. Give $E \oplus V$ the sum norm, define

$$Q_1 : E \oplus_1 V \rightarrow M, \quad Q_1(e, v) = (qe, v), \quad X = Z_{Q_1}, \quad Y = Z_q,$$

and put $L_q = \|q\|$, $K_q = \max\{L_q, 1\}$. Then X, Y are separable real Banach spaces and there is a bijection $\Psi : X \rightarrow Y$ such that

$$\text{Lip}(\Psi) \leq \max\{1 + K_q, b\}, \quad \text{Lip}(\Psi^{-1}) \leq \max\{1 + 2L_q/a, 1/a\}. \quad (3.1)$$

Moreover, X contains $c_0(V)$ isometrically and Y contains no closed subspace linearly isomorphic to $c_0(V)$.

Proof. The space E is separable because M is separable. The two operators Q_1 and q are bounded: in the sum norm,

$$\|Q_1(e, v)\|_M = (\|qe\|^2 + \|v\|^2)^{1/2} \leq K_q(\|e\| + \|v\|). \quad (3.2)$$

Thus Lemma 2.3 applies to X and Y .

Define the zero-preserving map

$$B : E \oplus_1 V \rightarrow E, \quad B(e, v) = e + \delta(qe, v). \quad (3.3)$$

The defining identity $q\delta(u, v) = h(u, v) - u$ gives

$$qB(e, v) = qe + h(qe, v) - qe = h(Q_1(e, v)). \quad (3.4)$$

For $z \in E$, the inverse of h is defined at qz ; set $x = h^{-1}(qz)$ and

$$C(z) = (z - \delta(x), \pi_V x). \quad (3.5)$$

Since

$$q(z - \delta(x)) = qz - (h(x) - \pi_U x) = \pi_U x,$$

we have $Q_1 C(z) = x$, and substitution in (3.3) gives $B(C(z)) = z$. Conversely, if $z = B(e, v)$, then (3.4) and injectivity of h give $h^{-1}(qz) = Q_1(e, v) = (qe, v)$. Formula (3.5) therefore returns (e, v) . Thus $C = B^{-1}$ and

$$Q_1 B^{-1} = h^{-1} q. \quad (3.6)$$

Only the inverse of h was used. In particular, this argument does not require q to be onto.

We next keep track of the two norms. For $w = (e, v)$ and $w' = (e', v')$, the isometry of δ and (3.2) yield

$$\|B(w) - B(w')\| \leq \|e - e'\| + \|Q_1(w - w')\| \leq (1 + K_q)\|w - w'\|_{E \oplus_1 V}.$$

For $z, z' \in E$, put $x = h^{-1}(qz)$, $x' = h^{-1}(qz')$. The lower bound for h implies

$$\|x - x'\|_M \leq \frac{L_q}{a}\|z - z'\|.$$

Using the *sum* norm in (3.5), rather than a Hilbert sum norm on $E \oplus V$, gives

$$\begin{aligned} \|B^{-1}(z) - B^{-1}(z')\|_{E \oplus_1 V} &\leq \|z - z'\| + \|\delta(x) - \delta(x')\| + \|\pi_V(x - x')\| \\ &\leq \|z - z'\| + 2\|x - x'\|_M \\ &\leq (1 + 2L_q/a)\|z - z'\|. \end{aligned}$$

Neither estimate divides by L_q ; they remain valid as written when $L_q = 0$. In the present setting $V = \ell_2 \neq 0$, that case cannot in fact occur under the hypotheses, since $q = 0$ would give $h(u, v) = u$, contradicting injectivity.

Define Ψ coordinatewise by

$$\Psi(s)_i = B(s_i) \quad (s \in X). \quad (3.7)$$

Because $B(0) = 0$, the bound for B makes $\sum_i 2^{-i}\|B(s_i)\|$ finite. Also $Q_1 s_i \rightarrow 0$ in M ; by (3.4) and continuity of h at zero, $qB(s_i) = h(Q_1 s_i) \rightarrow 0$. Hence $\Psi(s) \in Y$. Likewise $B^{-1}(0) = 0$ controls the weighted sum for the coordinatewise inverse, and (3.6) shows that if $qz_i \rightarrow 0$, then $Q_1 B^{-1}(z_i) = h^{-1}(qz_i) \rightarrow 0$. Thus the coordinatewise inverse maps Y into X , and the pointwise inverse identities make Ψ a bijection.

For clarity, the estimates apply to every difference of graph sequences. For $s, t \in X$,

$$\begin{aligned} \sum_i 2^{-i}\|B(s_i) - B(t_i)\| &\leq (1 + K_q) \sum_i 2^{-i}\|s_i - t_i\|, \\ \sup_i \|qB(s_i) - qB(t_i)\| &\leq b \sup_i \|Q_1(s_i - t_i)\|. \end{aligned}$$

Taking the maximum proves the first inequality in (3.1). For $z, z' \in Y$, the corresponding two bounds for B^{-1} and $Q_1 B^{-1} = h^{-1}q$ are

$$(1 + 2L_q/a) \sum_i 2^{-i}\|z_i - z'_i\|, \quad a^{-1} \sup_i \|q(z_i - z'_i)\|,$$

which prove the second. These are global two-point estimates, not only estimates at the origin.

Finally, define

$$\iota : c_0(V) \rightarrow X, \quad \iota((v_i)) = ((0, v_i))_i.$$

The sequence belongs to X , and

$$\|\iota((v_i))\|_X = \max \left\{ \sum_i 2^{-i}\|v_i\|, \sup_i \|v_i\| \right\} = \sup_i \|v_i\|,$$

because $\sum_{i \geq 1} 2^{-i} = 1$. This is a linear isometry. The Hilbert space M is superreflexive, so Theorem 2.2 makes $E = \mathcal{F}(M)$ weakly sequentially complete. By hypothesis q is completely continuous, and Lemma 2.5 therefore excludes a linear copy of $c_0(V)$ from Y . A linear isomorphism from X onto Y would carry the displayed closed copy to one in Y , which is impossible. $\square$

Remark 3.2. The identities $qB = hQ_1$ and $Q_1B^{-1} = h^{-1}q$ involve nonlinear maps on both sides. They do not exhibit a map $s : U \rightarrow E$ with $qs = \text{id}_U$, Lipschitz or otherwise. In particular, the quotient splitting corollary of Godefroy and Kalton [7, Corollary 3.2] cannot be invoked from these identities.

4 Local orthogonality forces complete continuity

The graph criterion requires complete continuity of a free-space linearization. The following theorem obtains it from a local coordinate condition on the original Lipschitz map. The neighborhoods are chosen around an arbitrary *finite family of distinct points*, and the orthogonality is required only after a common tail cutoff. Neither a global partition nor uniform tail decay on bounded sets is assumed.

Theorem 4.1 (Localized linearization). *Let $(M, \rho, 0)$ be a complete separable pointed metric space, and let*

$$U = \left(\bigoplus_{n \geq 1} U_n \right)_2$$

be a real Hilbert space with finite-dimensional blocks. Fix an orthonormal basis in every U_n , and write $\Pi_{\geq l}$ for projection onto $U_{\geq l} = (\bigoplus_{n \geq l} U_n)_2$. Suppose $D : M \rightarrow U$ is Lipschitz with $D(0) = 0$, and that for every $m \geq 1$ and every distinct $z_1, \dots, z_m \in M$ there are open neighborhoods O_j of z_j , an integer $l \geq 1$, and mutually orthogonal coordinate subspaces $H_j \subset U_{\geq l}$ such that

$$\Pi_{\geq l} D(O_j) \subset H_j \quad (1 \leq j \leq m). \quad (4.1)$$

A coordinate subspace means the closed span of a subset of the fixed orthonormal basis. Then

$$q = \widehat{D} : \mathcal{F}(M) \rightarrow U$$

is completely continuous.

The proof tests D against weakly null unit vectors. After passage to a subsequence, only finitely many points can retain local Lipschitz constants above a fixed positive threshold. We first prove an independent metric lemma that controls tests concentrated in shrinking balls around those points. Compact reduction will then control the remainder.

Lemma 4.2 (Tests in shrinking balls). *Let $(M, \rho, 0)$ be a pointed metric space, let $z \in M$, and let $\mu_i \rightarrow 0$ weakly in $\mathcal{F}(M)$. Suppose $r_i > 0$, $r_i \rightarrow 0$, and $f_i : M \rightarrow \mathbb{R}$ satisfy*

$$f_i(z) = 0, \quad \text{Lip}(f_i) \leq C, \quad f_i(x) = 0 \quad \text{when } \rho(x, z) \geq r_i,$$

where C does not depend on i . Then $\langle f_i, \mu_i \rangle \rightarrow 0$.

Proof. For $s > 0$, define

$$\phi_{z,s}(x) = \max\{0, \min\{1, 2 - 2\rho(x, z)/s\}\}.$$

This cutoff is one on $B(z, s/2)$, zero outside $B(z, s)$, takes values in $[0, 1]$, and has Lipschitz constant at most $2/s$. Whenever $f(z) = 0$ and $\text{Lip}(f) \leq C$, we claim

$$\text{Lip}(f\phi_{z,s}) \leq 3C, \quad \text{Lip}(f(1 - \phi_{z,s})) \leq 3C. \quad (4.2)$$

Let ψ be either cutoff in this claim. If $\psi(x) = \psi(y)$, the difference of the products is at most $C\rho(x, y)$. Otherwise at least one endpoint lies in $B(z, s)$; call it x . Then $|f(x)| \leq Cs$, and

$$f(x)\psi(x) - f(y)\psi(y) = \psi(y)(f(x) - f(y)) + f(x)(\psi(x) - \psi(y))$$

has absolute value at most $C\rho(x, y) + Cs(2/s)\rho(x, y) = 3C\rho(x, y)$. Thus (4.2) needs no global bound on f .

Suppose the conclusion fails. Passing to a subsequence, assume

$$|\langle f_i, \mu_i \rangle| \geq \epsilon > 0 \quad (i \geq 1).$$

Then $C > 0$. We recursively select increasing indices i_j , numbers $0 < s_j < r_{i_j}$, and functions

$$h_j = f_{i_j}(1 - \phi_{z, s_j})$$

such that

$$|\langle h_j, \mu_{i_j} \rangle| > \epsilon/2, \quad \text{Lip}(h_j) \leq 3C, \quad r_{i_{j+1}} < s_j/2. \quad (4.3)$$

After choosing i_j , approximate μ_{i_j} by a finite sum of evaluations

$$\nu_j = \sum_{k=1}^{N_j} a_{jk} \delta(x_{jk}), \quad \|\mu_{i_j} - \nu_j\| < \frac{\epsilon}{6C}.$$

Choose $s_j \in (0, r_{i_j})$ smaller than every positive distance from z to a member of $\{0, x_{j1}, \dots, x_{jN_j}\}$. If that finite set has no positive such distances, impose only $s_j < r_{i_j}$. The function $p_j = f_{i_j} \phi_{z, s_j}$ vanishes at every one of these points: outside the ball the cutoff is zero, and at a point equal to z the factor f_{i_j} is zero. Including the basepoint is essential when $z \neq 0$. It gives

$$\langle p_j, \nu_j \rangle = \sum_{k=1}^{N_j} a_{jk} (p_j(x_{jk}) - p_j(0)) = 0.$$

By (4.2),

$$|\langle p_j, \mu_{i_j} \rangle| \leq 3C \|\mu_{i_j} - \nu_j\| < \epsilon/2.$$

Subtracting from the original pairing proves the first inequality in (4.3); the second is (4.2). Finally choose $i_{j+1} > i_j$ so large that $r_{i_{j+1}} < s_j/2$. This completes the recursion.

The nonzero set of h_j lies in

$$\{x : s_j/2 < \rho(x, z) < r_{i_j}\}.$$

These annuli are pairwise disjoint. Every finite signed sum $H = \sum_{j \in A} \sigma_j h_j$, with $\sigma_j \in \{-1, 1\}$, therefore satisfies $\text{Lip}(H) \leq 6C$. Indeed, at a point at most one summand is nonzero. If two points lie in the same nonzero set, the $3C$ bound for that summand applies. If they lie in different nonzero sets, each of the two contributing functions vanishes at the other point, and their $3C$ bounds add. The cases where one or both points lie in no nonzero set are included. Subtracting $H(0)$ does not change this Lipschitz bound.

For every $\mu \in \mathcal{F}(M)$ and finite $A \subset \mathbb{N}$, real signs give

$$\sum_{j \in A} |\langle h_j, \mu \rangle| = \max_{\sigma_j \in \{-1, 1\}} \left| \left\langle \sum_{j \in A} \sigma_j h_j, \mu \right\rangle \right| \leq 6C \|\mu\|.$$

Taking increasing finite sets shows that

$$T : \mathcal{F}(M) \rightarrow \ell_1, \quad T\mu = (\langle h_j, \mu \rangle)_{j \geq 1}$$

is a bounded linear operator of norm at most $6C$. Thus $T\mu_{i_j} \rightarrow 0$ weakly in ℓ_1 . By the Schur property of ℓ_1 , its norms tend to zero. But its j th coordinate has absolute value greater than $\epsilon/2$ by (4.3), a contradiction. $\square$

Proof of Theorem 4.1. Let $E = \mathcal{F}(M)$ and $L = \text{Lip}(D) = \|q\|$. If $L = 0$, then $D = 0$ because $D(0) = 0$, and $q = 0$. Assume $L > 0$. If q were not completely continuous, after taking a subsequence there would be $\mu_i \rightarrow 0$ weakly in E and $\|q\mu_i\| \geq \epsilon_0 > 0$ for all i . Set

$$b_i = \frac{q\mu_i}{\|q\mu_i\|}, \quad d_i(x) = \langle D(x), b_i \rangle, \quad M_0 = \sup_i \|\mu_i\| < \infty.$$

The unit vectors b_i are weakly null. For each $b \in U$,

$$|\langle b_i, b \rangle| \leq \epsilon_0^{-1} |\langle q\mu_i, b \rangle| \rightarrow 0,$$

since a bounded linear map preserves weak convergence. Consequently

$$d_i(0) = 0, \quad \text{Lip}(d_i) \leq L, \quad \langle d_i, \mu_i \rangle = \|q\mu_i\| \geq \epsilon_0. \quad (4.4)$$

The functions d_i tend uniformly to zero on every compact subset of M . They tend pointwise to zero by weak nullity of b_i . Given a compact set and a positive error, take a finite sufficiently fine metric net; pointwise convergence on that net and the common Lipschitz bound control all points of the compact set.

Only finitely many points can retain a large local Lipschitz constant. Choose a countable base $\mathcal{B}$ for M , which exists by separability. Take the Lipschitz constant on a set of at most one point to be zero. For each $O \in \mathcal{B}$, $\text{Lip}(d_i|_O) \in [0, L]$. Successively take subsequences for the countably many O , then take a diagonal subsequence. Relabeling, we may assume that all limits

$$a_O = \lim_{i \rightarrow \infty} \text{Lip}(d_i|_O) \quad (O \in \mathcal{B})$$

exist. This subsequence is fixed before the next parameter is chosen.

Fix $\eta > 0$ and define

$$Z = \{z \in M : a_O \geq \eta \text{ for every } O \in \mathcal{B} \text{ containing } z\}.$$

Take any finite family of distinct $z_1, \dots, z_m \in Z$. Apply (4.1) and shrink the resulting neighborhoods to base neighborhoods O_j containing z_j . Let P_j be projection onto H_j , and put $\Pi_{< l} = I - \Pi_{\geq l}$. For $x, y \in O_j$, the tail $\Pi_{\geq l}(D(x) - D(y))$ lies in H_j . Hence

$$|d_i(x) - d_i(y)| \leq L\rho(x, y)(\|P_j b_i\| + \|\Pi_{< l} b_i\|).$$

The first $l - 1$ blocks are finite dimensional. Weak nullity of b_i makes each of its finitely many coordinates there tend to zero, so $\|\Pi_{< l} b_i\| \rightarrow 0$. Since $a_{O_j} \geq \eta$, the preceding inequality gives

$$\liminf_{i \rightarrow \infty} \|P_j b_i\| \geq \eta/L.$$

The P_j have mutually orthogonal ranges, and $\|b_i\| = 1$. It follows that

$$m \frac{\eta^2}{L^2} \leq \liminf_{i \rightarrow \infty} \sum_{j=1}^m \|P_j b_i\|^2 \leq 1.$$

The same conclusion holds for every finite family of distinct points of Z , so Z is finite and

$$|Z| \leq L^2/\eta^2. \quad (4.5)$$

Remove small neighborhoods of those finitely many points. Choose $r_0 > 0$ so that the balls $B(z, r)$, $z \in Z$, are pairwise disjoint for every $0 < r < r_0$. If Z has at most one point, any $r_0 > 0$ suffices. With the cutoffs used in Lemma 4.2, put

$$f_{i,z,r} = \phi_{z,r}(d_i - d_i(z)), \quad g_{i,r} = \sum_{z \in Z} f_{i,z,r}, \quad R_{i,r} = d_i - g_{i,r}.$$

An empty sum is zero. Equation (4.2) gives $\text{Lip}(f_{i,z,r}) \leq 3L$. Their nonzero sets lie in disjoint balls, so the two-point argument for finite signed sums in the proof of Lemma 4.2 gives

$$\text{Lip}(g_{i,r}) \leq 6L, \quad \text{Lip}(R_{i,r}) \leq 7L. \quad (4.6)$$

These constants do not depend on r or on $|Z|$.

For each fixed $z \in Z$, we have the ordered limit

$$\lim_{r \downarrow 0} \limsup_{i \rightarrow \infty} |\langle f_{i,z,r}, \mu_i \rangle| = 0. \quad (4.7)$$

If this failed, some $a > 0$ and radii $r_j \downarrow 0$ would satisfy $\limsup_i |\langle f_{i,z,r_j}, \mu_i \rangle| \geq 2a$. Choose increasing i_j with $|\langle f_{i_j,z,r_j}, \mu_{i_j} \rangle| \geq a$. These tests vanish at z , have Lipschitz constants at most $3L$, and vanish outside $B(z, r_j)$. The weakly null subsequence μ_{i_j} and Lemma 4.2 give a contradiction. Because Z is finite, the triangle inequality and (4.7) imply

$$\lim_{r \downarrow 0} \limsup_{i \rightarrow \infty} |\langle g_{i,r}, \mu_i \rangle| = 0. \quad (4.8)$$

If Z is empty, (4.8) is immediate.

The removed part is now controlled. It remains to show that the remainder has small pairing. This is where completeness of M enters through compact reduction.

Estimate the remainder on one compact set. Fix $\tau > 0$. By Theorem 2.1, choose a compact $K \subset M$, $0 \in K$, and $\nu_i \in \mathcal{F}(K)$ with

$$\|\mu_i - \nu_i\| < \tau, \quad \|\nu_i\| \leq M_0 + \tau.$$

The set K depends on τ and the sequence, but not on r . Now fix $0 < r < r_0$ while taking the sequence limit. Write $m = |Z|$ and

$$\Phi_r = \sum_{z \in Z} \phi_{z,r}, \quad \epsilon_i = \sup_{x \in K \cup Z} |d_i(x)|.$$

The compact set $K \cup Z$ contains 0, so $\epsilon_i \rightarrow 0$, including when Z is empty. Disjointness of the balls gives $0 \leq \Phi_r \leq 1$, and

$$R_{i,r} = (1 - \Phi_r)d_i + \sum_{z \in Z} \phi_{z,r}d_i(z). \quad (4.9)$$

For $x \in K$ the right side is a convex combination of values bounded in absolute value by ϵ_i . Therefore

$$\sup_{x \in K} |R_{i,r}(x)| \leq \epsilon_i. \quad (4.10)$$

On each inner ball $B(z, r/2)$, the remainder is the constant $d_i(z)$.

At any point $p \notin Z$, choose $O \in \mathcal{B}$ containing p with $a_O < \eta$. For $x, y \in K \cap O$, expansion of (4.9) gives

$$\begin{aligned} R_{i,r}(x) - R_{i,r}(y) &= (1 - \Phi_r(x))(d_i(x) - d_i(y)) \\ &\quad + \sum_{z \in Z} (\phi_{z,r}(x) - \phi_{z,r}(y))(d_i(z) - d_i(y)). \end{aligned}$$

The last factors have absolute value at most $2\epsilon_i$, and $\text{Lip}(\phi_{z,r}) \leq 2/r$. Thus

$$\text{Lip}(R_{i,r}|_{K \cap O}) \leq \text{Lip}(d_i|_O) + \frac{4m}{r}\epsilon_i. \quad (4.11)$$

The radius is fixed here, so the second term tends to zero with i .

The inner balls at points of Z , together with the chosen base neighborhoods at points outside Z , cover K . Choose a finite subcover $V_1, \dots, V_N$. The Lebesgue-number property supplies $\lambda > 0$ such that any pair of points of K at distance less than λ belongs to some one V_t . For an inner ball put $c_{i,t} = 0$; for a chosen base neighborhood put

$$c_{i,t} = \text{Lip}(d_i|_{V_t}) + (4m/r)\epsilon_i.$$

For pairs at distance less than λ , the Lipschitz quotient of $R_{i,r}$ is at most $\max_t c_{i,t}$, by constancy on inner balls and (4.11). For pairs at distance at least λ , (4.10) bounds it by $2\epsilon_i/\lambda$. Therefore

$$\text{Lip}(R_{i,r}|_K) \leq \max \left\{ \max_{1 \leq t \leq N} c_{i,t}, \frac{2\epsilon_i}{\lambda} \right\}.$$

If K has only one point, its Lipschitz constant is zero and the same conclusion holds. There are finitely many V_t , and for each base neighborhood $a_{V_t} < \eta$. Taking the upper limit gives

$$\limsup_{i \rightarrow \infty} \text{Lip}(R_{i,r}|_K) \leq \eta. \quad (4.12)$$

Use (2.2) for $\nu_i \in \mathcal{F}(K)$, and use the global bound (4.6) for the approximation error. We obtain

$$\begin{aligned} |\langle R_{i,r}, \mu_i \rangle| &\leq |\langle R_{i,r}, \nu_i \rangle| + |\langle R_{i,r}, \mu_i - \nu_i \rangle| \\ &\leq (M_0 + \tau) \text{Lip}(R_{i,r}|_K) + 7L\tau. \end{aligned}$$

With η, τ, K, r fixed, (4.12) implies

$$\limsup_{i \rightarrow \infty} |\langle R_{i,r}, \mu_i \rangle| \leq \eta(M_0 + \tau) + 7L\tau.$$

The right side is independent of r . Since $d_i = R_{i,r} + g_{i,r}$, we may now let $r \downarrow 0$ and use (4.8) to get

$$\limsup_{i \rightarrow \infty} |\langle d_i, \mu_i \rangle| \leq \eta(M_0 + \tau) + 7L\tau \quad \text{for every } \eta > 0, \tau > 0. \quad (4.13)$$

The quantifier order is as follows: the diagonal subsequence is fixed first; for each η choose Z, r_0 ; for each τ choose K ; for a fixed $r < r_0$ take $i \rightarrow \infty$; then take $r \downarrow 0, \tau \downarrow 0$, and finally $\eta \downarrow 0$. No uniform convergence in both i and r has been used. The last two limits in (4.13) make its right side zero, contradicting (4.4). $\square$

5 A localized change of variables in Hilbert space

We now construct the Hilbert-space map required by Proposition 3.1. The main issue is to let the output coordinates separate input neighborhoods without losing global invertibility.

Theorem 5.1. *There are finite-dimensional Euclidean coordinate spaces U_n , $U = (\bigoplus_{n \geq 1} U_n)_2$, and a bijection $h : U \oplus_2 \ell_2 \rightarrow U$ fixing zero, such that*

$$\frac{24}{25} \|x - x'\| \leq \|h(x) - h(x')\| \leq \frac{26}{25} \|x - x'\|. \quad (5.1)$$

Moreover, $D = h - \pi_U$ has local orthogonality: for every finite family of distinct points $x_1, \dots, x_m$ there are neighborhoods O_j of x_j , an integer l , and mutually orthogonal coordinate subspaces $H_j \subset (\bigoplus_{n \geq l} U_n)_2$ such that

$$P_{\geq l} D(O_j) \subset H_j \quad (1 \leq j \leq m).$$

Here $P_{\geq l}$ is the projection onto blocks with index at least l , and a coordinate subspace is the closed span of a subset of the specified orthonormal coordinates.

We begin with an onto isometric rearrangement determined by one unit vector in each block. Allowing these vectors to depend on the input will produce h . The rearrangement identifies the two requirements on their motion: a weighted derivative bound will control metric distortion, and separation of their coordinate supports will give local orthogonality. We then construct the vectors, prove global invertibility of finite approximants, and use an inverse-tail estimate to show that their limiting map is onto.

5.1 The frozen rearrangement

For the moment let $U = (\bigoplus_{n \geq 1} U_n)_2$ have arbitrary nonzero finite-dimensional real Hilbert blocks, let $V = \ell_2$, and put $M = U \oplus_2 V$. Choose a unit vector $\xi_n \in U_n$ for each n . For $x = (u, v) \in M$, write

$$\alpha_n = \langle \xi_n, u_n \rangle, \quad u_n^\perp = u_n - \alpha_n \xi_n, \quad C_{2m} = \alpha_m, \quad C_{2m-1} = v_m.$$

Define

$$(L_\xi x)_n = u_n^\perp + C_n \xi_n. \quad (5.2)$$

Thus the perpendicular component stays in block n ; the old slot coefficient α_m moves to the slot in block $2m$, and v_m fills the slot in block $2m - 1$. Figure 1 shows this rearrangement.

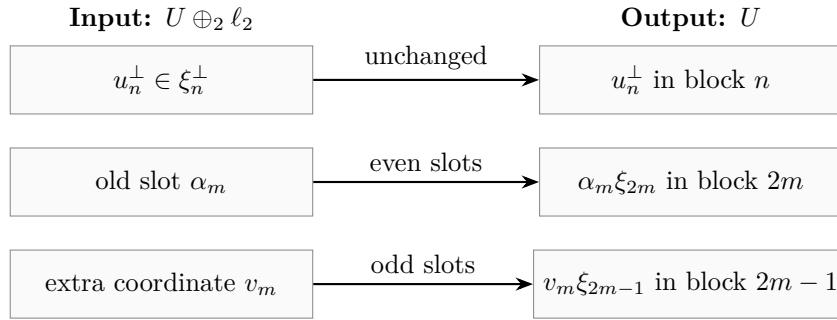


Figure 1: The frozen map L_ξ . The perpendicular components and the two scalar sequences form orthogonal coordinates. The picture describes the frozen linear isometry; when the slots depend on the input, it gives the pointwise norm identity but not preservation of distances between distinct inputs.

Lemma 5.2. $L_\xi : M \rightarrow U$ is an onto linear isometry. If $y = (y_n) \in U$ and $\beta_n = \langle \xi_n, y_n \rangle$, its inverse is

$$(L_\xi^{-1}y)_{U_m} = y_m - \beta_m \xi_m + \beta_{2m} \xi_m, \quad (L_\xi^{-1}y)_{V_m} = \beta_{2m-1}. \quad (5.3)$$

Proof. All formulas are linear for fixed ξ . Since $u_n^\perp \perp \xi_n$, the squared norm of (5.2) is

$$\begin{aligned} \|L_\xi x\|^2 &= \sum_n \|u_n^\perp\|^2 + \sum_n |C_n|^2 \\ &= \sum_n (\|u_n\|^2 - |\alpha_n|^2) + \sum_m |\alpha_m|^2 + \sum_m |v_m|^2 = \|x\|_M^2. \end{aligned}$$

This also proves the output belongs to U .

For $y \in U$, $|\beta_n| \leq \|y_n\|$, so $(\beta_n) \in \ell_2$. The first two terms $y_m - \beta_m \xi_m$ in (5.3) are perpendicular to its last term. The prescribed vector therefore belongs to M , since

$$\begin{aligned} &\sum_m \left(\|y_m - \beta_m \xi_m + \beta_{2m} \xi_m\|^2 + |\beta_{2m-1}|^2 \right) \\ &= \sum_m (\|y_m\|^2 - |\beta_m|^2) + \sum_m |\beta_{2m}|^2 + \sum_m |\beta_{2m-1}|^2 = \|y\|^2. \end{aligned}$$

For this input the old slot coefficient is β_{2m} , and the perpendicular component in block m is $y_m - \beta_m \xi_m$. The even and odd output slots consequently recover every β_n , so (5.2) returns y . This proves surjectivity and the inverse formula. $\square$

5.2 What the moving slots must satisfy

Given continuously differentiable unit-vector maps $w_n : M \rightarrow U_n$, define $h(x) = L_{w(x)}x$. Explicitly,

$$\begin{aligned} h(x)_n &= u_n + w_n(x)(C_n(x) - \langle w_n(x), u_n \rangle), \\ C_{2m}(x) &= \langle w_m(x), u_m \rangle, \quad C_{2m-1}(x) = v_m. \end{aligned} \quad (5.4)$$

The frozen isometry shows that this is well-defined and gives

$$\|h(x)\| = \|x\|. \quad (5.5)$$

This identity does not yet compare distinct inputs, whose slots may differ. To control that difference, fix $c = 1/100$ and define the nonnegative weights

$$a_n(x)^2 = \|u_n\|^2 + \|u_{\lceil n/2 \rceil}\|^2 + |v_{\lceil n/2 \rceil}|^2. \quad (5.6)$$

These weights follow the routing in (5.4): $|C_n(x)| \leq a_n(x)$ and $\|u_n\| \leq a_n(x)$, so the scalar coefficient multiplying $w_n(x)$ has absolute value at most $2a_n(x)$. We shall choose the slots so that

$$\sum_{n \geq 1} a_n(x)^2 \|Dw_n(x)\|^2 \leq c^2 \quad (x \in M). \quad (5.7)$$

We shall show that each derivative of a finite approximant is within $4c$ of a frozen isometry.

The second requirement concerns the coordinate supports. For every finite family of distinct input points, we shall find neighborhoods whose slots use disjoint coordinates in all sufficiently high blocks. Indeed, the n th block of $D(x) = h(x) - u$ is a scalar multiple of $w_n(x)$, so this separation gives the required local orthogonality of D . We now choose the block dimensions and construct slots satisfying both requirements.

5.3 Finite stages and separating coordinates

Choose a nonnegative smooth function $\psi : \mathbb{R} \rightarrow \mathbb{R}$ that is positive on $(-1, 1)$ and zero outside $[-1, 1]$. For $k \geq 1$, put

$$\mathcal{A}_k = \{-k, -k+1, \dots, k\}^k.$$

For $a = (a_1, \dots, a_k) \in \mathcal{A}_k$, define

$$G_{k,a}(t) = \prod_{j=1}^k \psi(kt_j - a_j) \quad (t \in [-1, 1]^k).$$

Every point of $[-1, 1]$ is within $1/(2k)$ of some a_j/k with $-k \leq a_j \leq k$. Thus $G_k(t) = (G_{k,a}(t))_{a \in \mathcal{A}_k}$ is never zero on the cube. Its Euclidean norm has a positive minimum there. Each factor $t \mapsto \psi(kt - a_j)$ is supported on an interval of length $2/k$.

Give U_n an orthonormal basis consisting of a single stage-zero vector and a separate set of coordinates indexed by $\mathcal{A}_k$ for each $1 \leq k \leq n$. Different stages are orthogonal. In particular, we may take

$$\dim U_n = 1 + \sum_{k=1}^n (2k+1)^k. \quad (5.8)$$

From now on, use these blocks in the separable Hilbert spaces

$$U = \left(\bigoplus_{n \geq 1} U_n \right)_2, \quad V = \ell_2, \quad M = U \oplus_2 V$$

on which the slots will be constructed. Enumerate the scalar coordinates of M and apply $\tanh$ to each of them. This gives smooth Lipschitz functions $p_j : M \rightarrow [-1, 1]$ that separate points.

Let $W_{n,0}$ be the constant stage-zero unit vector. For $1 \leq k \leq n$, let $W_{n,k}(x)$ be the vector supported on stage k with coordinates

$$W_{n,k}(x)_a = \frac{G_{k,a}(p_1(x), \dots, p_k(x))}{\|G_k(p_1(x), \dots, p_k(x))\|_2}. \quad (5.9)$$

It is a smooth unit-vector map. Compactness of the cube, the positive lower bound for the denominator, and boundedness of the derivatives of the p_j give

$$\sup_x \|DW_{n,k}(x)\| \leq L_k < \infty.$$

The same L_k works for every $n \geq k$, since their stage- k coordinate formulas are identical. Put $L_0 = 0$. No bound uniform in k is needed.

5.4 A common budget for motion

To obtain (5.7), use the tails of the weights (5.6) as nonnegative radii:

$$r_n(x)^2 = \sum_{j \geq n} a_j(x)^2.$$

Every input coordinate in the second and third terms of (5.6) is counted twice, so

$$r_1(x)^2 = 3\|u\|^2 + 2\|v\|^2, \quad a_n(x)^2 = r_n(x)^2 - r_{n+1}(x)^2. \quad (5.10)$$

Each r_n is the norm of a bounded linear map into a Hilbert space. It is $\sqrt{3}$ -Lipschitz, smooth where positive, and tends to zero at every fixed x .

With the fixed constant $c = 1/100$, set

$$\gamma(r) = \begin{cases} \frac{c}{r \log(1/r)}, & 0 < r < e^{-2}, \\ 0, & r \geq e^{-2}. \end{cases} \quad (5.11)$$

This function is nonincreasing as r increases. It has an infinite first integral near zero but a finite weighted square integral:

$$\int_0^b \gamma(r) dr = \infty \quad (b > 0), \quad \int_0^\infty 2r\gamma(r)^2 dr = 2c^2 \int_2^\infty t^{-2} dt = c^2. \quad (5.12)$$

The infinite first integral permits arbitrarily many rotations; the second estimate will bound their total effect on the derivative.

Choose successive compact intervals $[a'_k, b'_k] \subset (0, e^{-2})$ for the transitions from stage $k-1$ to stage k , with

$$b'_{k+1} < a'_k, \quad b'_k < 2^{-k}, \quad \gamma(r)/2 \geq L_{k-1} + L_k \quad (0 < r \leq b'_k).$$

They can be chosen together with smooth nonincreasing angles $\theta_k : [0, \infty) \rightarrow [0, \pi/2]$ such that

$$\theta_k(r) = \pi/2 \ (r \leq a'_k), \quad \theta_k(r) = 0 \ (r \geq b'_k), \quad |\theta'_k(r)| \leq \gamma(r)/(2\sqrt{3}), \quad (5.13)$$

and the angles are constant near the interval endpoints. Here is a direct construction. First choose b'_k small enough for the derivative bound and the preceding interval. By (5.12), a sufficiently small $a'_k > 0$ makes the integral of $\gamma/(2\sqrt{3})$ over (a'_k, b'_k) larger than $\pi/2$. A smooth nonnegative function supported inside this interval, bounded above by $\gamma/(2\sqrt{3})$ and with integral larger than $\pi/2$, exists by approximation from below. Rescale it to integral $\pi/2$ and integrate from r to $+\infty$ to obtain θ_k .

For each n , define a unit-vector map $P_n(x, r)$ by following these same timings through stages $0, \dots, n$ and then stopping. On the k th transition interval, $k \leq n$, put

$$P_n(x, r) = \cos \theta_k(r) W_{n, k-1}(x) + \sin \theta_k(r) W_{n, k}(x).$$

Between intervals use the intervening single stage; use $W_{n, 0}$ above b'_1 and $W_{n, n}$ below a'_n , including at $r = 0$. Orthogonality of the stages makes this a unit vector. The endpoint choices make P_n continuously differentiable. On a transition, (5.13) and the bounds for L_k give

$$\|\partial_x P_n(x, r)\| \leq \gamma(r)/2, \quad \|\partial_r P_n(x, r)\| \leq \gamma(r)/(2\sqrt{3}) \quad (r > 0). \quad (5.14)$$

Between transitions the first bound follows from monotonicity of γ , and the second derivative is zero. Stage zero is constant, so both bounds also hold where $\gamma = 0$.

Set

$$w_n(x) = P_n(x, r_n(x)). \quad (5.15)$$

Although r_n need not be differentiable where it vanishes, w_n is continuously differentiable there: in a neighborhood of any such point, $r_n < a'_n$ and $w_n = W_{n, n}$. At a point with $r_n > 0$, the chain rule and (5.14) give $\|Dw_n(x)\| \leq \gamma(r_n(x))$. These slots satisfy the required budget (5.7). Indeed, when $r_n > 0$, monotonicity gives

$$(r_n^2 - r_{n+1}^2)\gamma(r_n)^2 \leq \int_{r_{n+1}}^{r_n} 2r\gamma(r)^2 dr,$$

and summing uses (5.12). When $r_n = 0$, also $a_n = 0$, so that term in (5.7) is zero.

The derivative budget controls metric distortion. We still need the independent locality property that will make the free-space linearization completely continuous.

5.5 Local orthogonality of the slots

Lemma 5.3. *For every finite family of distinct points of M there are neighborhoods O_j and an integer l such that the coordinate supports of $\{w_n(x) : x \in O_j, n \geq l\}$ are disjoint for different j .*

Proof. For an empty family, take $l = 1$ and the empty family of neighborhoods. For a singleton, take $l = 1$ and $O_1 = M$. In either case there is no disjointness condition to check. Now consider two distinct points. Some p_j has different values at them. Choose $k_0 \geq j$ sufficiently large and neighborhoods on which all values of that p_j on the two sides are separated by more than $2/k_0$. At any stage $k \geq k_0$, a coordinate label that occurs on both sides would require the j th factor $\psi(kp_j - a_j)$ to be nonzero at points x, y on opposite sides. Then both $p_j(x)$ and $p_j(y)$ would lie in $((a_j - 1)/k, (a_j + 1)/k)$, contradicting $|p_j(x) - p_j(y)| > 2/k_0 \geq 2/k$. Thus the unions of the supports on the two neighborhoods are disjoint at every common stage $k \geq k_0$.

Fix $0 < \rho < a'_{k_0}$, below the completed transition into stage k_0 . Choose $l \geq k_0$ so that r_l at each center is less than $\rho/2$, and shrink both neighborhoods to have radius less than $\rho/(2\sqrt{3})$. Since $r_n \leq r_l$ for $n \geq l$ and r_l is $\sqrt{3}$ -Lipschitz, we have $r_n(x) < \rho$ throughout either neighborhood whenever $n \geq l$. Only stages $k \geq k_0$ are then active in $w_n(x)$. Distinct stages are orthogonal, and supports at the same stage are disjoint by the first paragraph. This proves the two-point assertion. For finitely many points, intersect the neighborhoods obtained for every pair and take the largest cutoff. $\square$

The use of small neighborhoods here is essential: pointwise decay of the r_n and their common Lipschitz bound suffice. Uniform decay on bounded subsets of M is neither claimed nor needed.

The constructed slots now satisfy both the derivative budget and local separation. Use them in (5.4) to define h . It remains to prove the two-point bounds and surjectivity; the pointwise norm identity alone does not give either conclusion.

5.6 Finite approximants and global inverses

Let h_N be the same rearrangement with slots

$$w_n^{(N)}(x) = \begin{cases} w_n(x), & n \leq N, \\ W_{n,0}, & n > N. \end{cases}$$

Let A_0 denote the isometry with all slots equal to $W_{n,0}$. Only output blocks $n \leq 2N$ of $h_N - A_0$ can be nonzero, and their formulas are continuously differentiable. Thus h_N is a continuously differentiable map between Hilbert spaces. It also satisfies $\|h_N(x)\| = \|x\|$.

Lemma 5.4. *Each $h_N : M \rightarrow U$ is a continuously differentiable bijection with continuously differentiable inverse, and*

$$(1 - 4c)\|x - x'\| \leq \|h_N(x) - h_N(x')\| \leq (1 + 4c)\|x - x'\|. \quad (5.16)$$

Proof. Fix a point and a unit direction, and abbreviate the approximant slots $w_n^{(N)}(x)$ and their directional derivatives by $w_n, \dot{w}_n$; the slots with $n > N$ have derivative zero. Differentiating the inputs in (5.4) with the slots frozen gives the onto isometry $L_{w^{(N)}(x)}$. The remaining terms in output block n are

$$\dot{w}_n(C_n - \langle w_n, u_n \rangle) + w_n(\dot{C}_n - \langle \dot{w}_n, u_n \rangle), \quad \dot{C}_{2m} = \langle \dot{w}_m, u_m \rangle, \quad \dot{C}_{2m-1} = 0.$$

Put $R^2 = \sum_n a_n(x)^2 \|\dot{w}_n\|^2 \leq c^2$. Since $|C_n - \langle w_n, u_n \rangle| \leq 2a_n$, the first term has Hilbert norm at most $2R$. Each of the scalar sequences $(\dot{C}_n)_n$ and $(\langle \dot{w}_n, u_n \rangle)_n$ has ℓ_2 norm at most R , because $a_n \geq \|u_n\|$. Therefore

$$\|Dh_N(x) - L_{w^{(N)}(x)}\| \leq 4c. \quad (5.17)$$

A Neumann-series perturbation of the isometry gives

$$\|Dh_N(x)\| \leq 1 + 4c, \quad \|Dh_N(x)^{-1}\| \leq (1 - 4c)^{-1}. \quad (5.18)$$

We next establish global bijectivity before using the lower Lipschitz estimate. Put $F_N = \bigoplus_{n \leq 2N} U_n$ and let P project onto $F_N^\perp$. We have

$$Ph_N = PA_0.$$

For a fixed $b \in F_N^\perp$, the map h_N thus restricts to

$$A_0^{-1}(b) + A_0^{-1}(F_N) \longrightarrow b + F_N. \quad (5.19)$$

These affine spaces have the same finite dimension. The derivative is an isomorphism between their tangent spaces. Indeed it maps $A_0^{-1}(F_N)$ into F_N ; conversely, for $z \in F_N$, the vector $t = Dh_N(x)^{-1}z$ satisfies $PA_0t = PDh_N(x)t = 0$. Thus (5.19) is a local diffeomorphism. It is proper, since an inverse image of a compact set is closed and norm-bounded by $\|h_N(x)\| = \|x\|$, hence compact in this finite-dimensional source fiber.

A proper local diffeomorphism between affine Euclidean spaces of the same dimension is bijective. To recall the topological argument, its image is open by local invertibility and closed by properness, so it is the entire target. Each fiber is compact and discrete, hence finite. Local inverse neighborhoods, together with properness to exclude further preimages nearby, make it a covering map. A covering of a simply connected Euclidean space with connected source has one sheet. Applying this to (5.19) proves that h_N is bijective on each fiber and therefore on all of M .

The local inverses supplied by the inverse function theorem agree and form a global continuously differentiable inverse. Integrating the two bounds in (5.18) along line segments, for h_N and for its now-defined inverse, proves (5.16). $\square$

5.7 Passing to an onto limit

Completion of the proof of Theorem 5.1. For fixed $x = (u, v)$, the first N output blocks of $h_N(x)$ and $h(x)$ agree. Indeed, their own slots agree in those blocks, and the old slot used in an even block $2m \leq N$ has $m \leq N$. For either list $\zeta = w(x)$ or $\zeta = w^{(N)}(x)$, write

$$C_{2m}^\zeta = \langle \zeta_m, u_m \rangle, \quad C_{2m-1}^\zeta = v_m.$$

Then $(L_\zeta x)_n - u_n = \zeta_n(C_n^\zeta - \langle \zeta_n, u_n \rangle)$. Since $|C_n^\zeta| \leq a_n(x)$ and $\|u_n\| \leq a_n(x)$, each difference from u_n has norm at most $2a_n(x)$. For the remaining blocks, the difference between the two maps has norm at most $4a_n(x)$. Hence

$$\|h_N(x) - h(x)\| \leq 4r_{N+1}(x) \longrightarrow 0. \quad (5.20)$$

For fixed x, x' , take $N \rightarrow \infty$ in (5.16). This proves the two-sided bounds (5.1), since $1 - 4c = 24/25$ and $1 + 4c = 26/25$. In particular, h is injective. It fixes zero by (5.5).

We now prove surjectivity. Fix $y = (y_n) \in U$ and, using Lemma 5.4, put

$$x_N = h_N^{-1}(y).$$

Freeze the slots of h_N at this particular input:

$$\xi_n^{(N)} = w_n^{(N)}(x_N), \quad \beta_n^{(N)} = \langle \xi_n^{(N)}, y_n \rangle.$$

The directions may vary with N . Nevertheless $y = L_{\xi^{(N)}}x_N$ is an exact equality for each N , so Lemma 5.2 gives

$$(x_N)_{U_m} = y_m - \beta_m^{(N)}\xi_m^{(N)} + \beta_{2m}^{(N)}\xi_m^{(N)}, \quad (x_N)_{V_m} = \beta_{2m-1}^{(N)}. \quad (5.21)$$

The component $y_m - \beta_m^{(N)}\xi_m^{(N)}$ is perpendicular to $\xi_m^{(N)}$. Thus its squared norm is $\|y_m\|^2 - |\beta_m^{(N)}|^2$, and the last term in the U_m formula adds orthogonally.

For every integer $s \geq 1$, it follows that

$$\begin{aligned}
& \sum_{m \geq s} \left(\|(x_N)_{U_m}\|^2 + |(x_N)_{V_m}|^2 \right) \\
&= \sum_{m \geq s} \left(\|y_m\|^2 - |\beta_m^{(N)}|^2 + |\beta_{2m}^{(N)}|^2 + |\beta_{2m-1}^{(N)}|^2 \right) \\
&= \sum_{n \geq s} \|y_n\|^2 - \sum_{n=s}^{2s-2} |\beta_n^{(N)}|^2 \leq \sum_{n \geq s} \|y_n\|^2.
\end{aligned} \tag{5.22}$$

Here $|\beta_n^{(N)}| \leq \|y_n\|$, so all squared-coefficient series converge. The last equality is exact: the even and odd indices $2m, 2m-1$ for $m \geq s$ partition the integers $n \geq 2s-1$; their sum cancels that tail of $\sum_{n \geq s} |\beta_n^{(N)}|^2$. The only uncanceled negative terms have $s \leq n \leq 2s-2$. This finite sum is empty when $s = 1$. Keeping it is what avoids double counting the output's perpendicular and slot components.

The right side of (5.22) tends to zero with s , independently of N . At $s = 1$ the equality gives $\|x_N\| = \|y\|$. For each s , the heads

$$((x_N)_{U_m}, (x_N)_{V_m})_{m < s}$$

lie in a bounded subset of the finite-dimensional space $\bigoplus_{m < s} (U_m \oplus \mathbb{R})$. By successive subsequence selection in these finite-dimensional spaces and a diagonal choice, there is an increasing sequence N_j for which every fixed head converges.

This diagonal subsequence is Cauchy in the full norm. Given $\epsilon > 0$, choose s so that $\sum_{n \geq s} \|y_n\|^2 < (\epsilon/4)^2$. By (5.22), the tail of each of x_{N_j} and x_{N_k} has norm less than $\epsilon/4$. For sufficiently large j, k , convergence of the fixed head makes its difference smaller than $\epsilon/2$. The triangle inequality then gives $\|x_{N_j} - x_{N_k}\| < \epsilon$. Completeness of M yields $x_{N_j} \rightarrow x$ in norm for some $x \in M$.

Finally, apply the pointwise convergence estimate at this fixed limit x , and use the common upper Lipschitz bound for h_{N_j} :

$$\begin{aligned}
\|h(x) - y\| &\leq \|h(x) - h_{N_j}(x)\| + \|h_{N_j}(x) - h_{N_j}(x_{N_j})\| \\
&\leq 4r_{N_j+1}(x) + (1 + 4c)\|x - x_{N_j}\| \rightarrow 0.
\end{aligned}$$

Thus $h(x) = y$. Since y was arbitrary, h is onto.

The local orthogonality asserted in the theorem follows from Lemma 5.3. For its neighborhoods O_j and cutoff l , let H_j be the closed span of the union of coordinate supports of $w_n(x)$ with $x \in O_j$ and $n \geq l$. These are mutually orthogonal coordinate subspaces of $U_{\geq l}$. Each block $D(x)_n$ is a scalar multiple of $w_n(x)$ by (5.4). Its convergent tail therefore belongs to H_j for every $x \in O_j$, because H_j is closed. This verifies exactly (4.1) and completes the proof. $\square$

6 The counterexample

The Hilbert construction supplies the geometric hypotheses of localized linearization. The resulting completely continuous operator then meets the weighted graph criterion.

Proof of Theorem 1.1. Take the real Hilbert spaces $U, V = \ell_2, M = U \oplus_2 V$ and the map $h : M \rightarrow U$ from Theorem 5.1. The blocks U_n are finite dimensional, so U and M are separable; as Hilbert spaces they are complete. Put

$$E = \mathcal{F}(M), \quad D = h - \pi_U, \quad q = \hat{D} : E \rightarrow U.$$

The map D is Lipschitz, $D(0) = 0$, and it has the finite-family tail-coordinate orthogonality required by Theorem 4.1. That theorem makes q completely continuous. By (2.3) and (5.1),

$$\|q\| = \text{Lip}(D) \leq \text{Lip}(h) + \|\pi_U\| \leq \frac{26}{25} + 1 = \frac{51}{25}.$$

Set

$$Q_1 : E \oplus_1 \ell_2 \rightarrow M, \quad Q_1(e, v) = (qe, v), \quad X = Z_{Q_1}, \quad Y = Z_q.$$

All hypotheses of Proposition 3.1 hold with $a = 24/25$, $b = 26/25$. It supplies the coordinate-wise bijection $\Psi : X \rightarrow Y$, an isometric copy of $c_0(\ell_2)$ in X , and exclusion of every linearly isomorphic copy from Y . In particular X and Y are not linearly isomorphic.

Since $L_q \leq 51/25$ and $K_q = \max\{L_q, 1\} \leq 51/25$, (3.1) gives

$$\begin{aligned} \text{Lip}(\Psi) &\leq \max \left\{ 1 + \frac{51}{25}, \frac{26}{25} \right\} = \frac{76}{25}, \\ \text{Lip}(\Psi^{-1}) &\leq \max \left\{ 1 + \frac{2(51/25)}{24/25}, \frac{25}{24} \right\} = \frac{21}{4}. \end{aligned}$$

The latter inequality is equivalent to the lower bound $\|\Psi(s) - \Psi(t)\|_Y \geq (4/21)\|s - t\|_X$. Both bounds hold for all pairs $s, t \in X$, exactly as in (1.1). $\square$

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